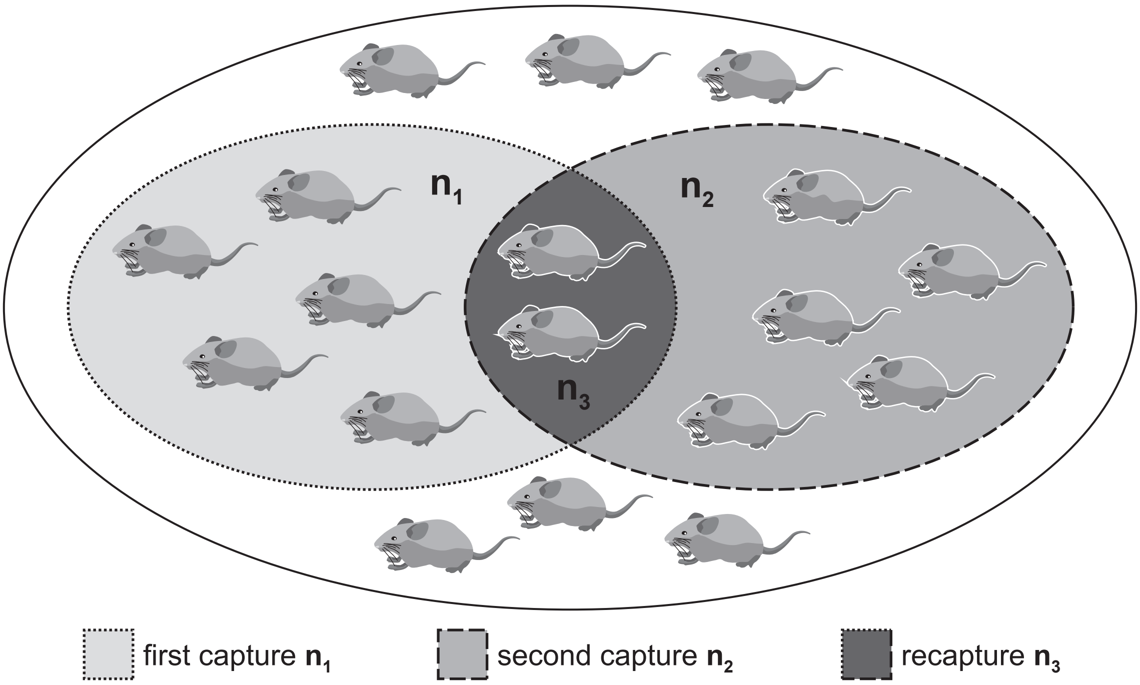
Examiner only

7. A pest control officer was asked to eradicate mice from a small food warehouse. By looking at the amount of food damage, she made an initial, on-the-spot, estimate of the number of mice in the warehouse. This initial estimate made by the pest control officer is the total number of mice shown in **Image 7.1**.

Image 7.1



In order to get a better estimate of the size of the mouse population in the warehouse the pest control officer used the capture/recapture technique. She set live-capture traps in the warehouse. On the next day the traps were inspected, all the mice caught were counted, marked and released. The mice caught in this first capture are represented by (**n_1**) in **Image 7.1**.

Two days later the traps were reset. The next day the traps were again inspected. The mice caught in this second capture are represented by (**n_2**). Within the second capture the number of marked mice were counted. These marked, recaptured, mice are represented by (**n_3**) in **Image 7.1**.

(a) (i) Use the following equation to estimate the population size of the mice in the warehouse. Give your answer to the nearest whole number. [2]

$$\text{population size} = \frac{\text{number in 1st capture} \times \text{number in 2nd capture}}{\text{number in 2nd capture previously marked}}$$

Population size = ...

